

Lecture 10: Partial Derivatives

Background & Motivation

Now we differentiate. The idea is simple: hold all variables constant except one, and differentiate with respect to that one. This gives us partial derivatives—the multivariate analog of $f'(x)$. Today we define them, compute them, and see how they measure the slope of a surface in each coordinate direction.

1. Partial Derivatives

Definition 1: Partial Derivative (Limit Definition)

The **partial derivative** $f_x(a, b)$ is the instantaneous rate of change of $f(x, y)$ in the x -direction at the point (a, b) , with y held fixed at b . Geometrically, slice the graph of f with the vertical plane $y = b$; the resulting cross-section is a single-variable curve $z = f(x, b)$, and $f_x(a, b)$ is the **slope of the tangent line** to that curve at $x = a$. In other words, partial derivatives are ordinary single-variable slopes taken along axis-aligned slices of the surface.

$$f_x(x, y) = \frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}$$

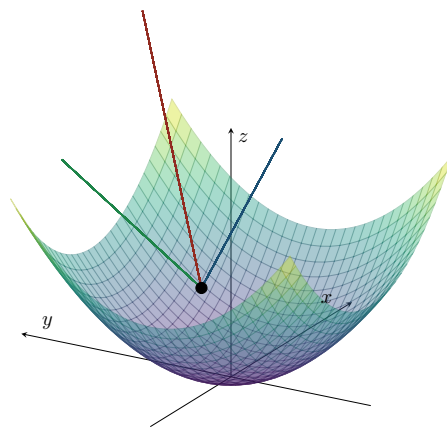
With respect to y :

$$f_y(x, y) = \frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Practical rule: To find f_x , treat y as a constant and differentiate with respect to x . For f_y , treat x as constant.

Geometric Interpretation

- $f_x(a, b)$ is the slope of the tangent line to $z = f(x, b)$ at $x = a$
- Measures rate of change of f in the x -direction while y is held fixed
- Similarly, $f_y(a, b)$ is the slope of $z = f(a, y)$ at $y = b$



Tangent line (red) at $(1, 1, 2)$ on $z = x^2 + y^2$. Blue dashed: xz -projection (slope $= f_x = 2$). Green dashed: yz -projection (slope $= f_y = 2$).

Example 1: Using the Limit Definition

Use the limit definition to find f_x for $f(x, y) = x^2y + 3xy^2$.

1. **Set up:** $f_x = \lim_{h \rightarrow 0} \frac{(x+h)^2y + 3(x+h)y^2 - x^2y - 3xy^2}{h}$

2. **Expand:** $= \lim_{h \rightarrow 0} \frac{x^2y + 2xhy + h^2y + 3xy^2 + 3hy^2 - x^2y - 3xy^2}{h}$
 $= \lim_{h \rightarrow 0} \frac{2xhy + h^2y + 3hy^2}{h}$

3. **Simplify and evaluate:** $= \lim_{h \rightarrow 0} (2xy + hy + 3y^2) = \boxed{2xy + 3y^2}$

Example 2: Computing Partial Derivatives

Find all first partial derivatives:

(a) $f(x, y) = x^3e^y + \sin(xy)$

$$f_x = 3x^2e^y + y \cos(xy)$$

$$f_y = x^3e^y + x \cos(xy)$$

(b) $g(x, y) = \frac{x^2}{x + y}$

$$g_x = \frac{2x(x+y) - x^2}{(x+y)^2} = \frac{x^2 + 2xy}{(x+y)^2} \quad (\text{quotient rule in } x)$$

$$g_y = \frac{0 - x^2}{(x+y)^2} = \frac{-x^2}{(x+y)^2}$$

(c) $h(x, y, z) = xye^z + z \ln(x + y)$

$$h_x = ye^z + \frac{z}{x+y}, \quad h_y = xe^z + \frac{z}{x+y}, \quad h_z = xye^z + \ln(x + y)$$

2. Derivative Rules for Partialials

Rules (Hold Other Variables Constant)

- **Linearity:** $\partial_x[af + bg] = af_x + bg_x$
- **Product:** $\partial_x[fg] = f_xg + fg_x$
- **Quotient:** $\partial_x[f/g] = (f_xg - fg_x)/g^2$
- **Chain:** $\partial_x[f(g(x, y))] = f'(g) \cdot g_x$
- **Zero partial:** if $f_x = 0$ everywhere $\implies f$ does not depend on x
- **Constants:** $\partial_x[c] = 0$; treat other variables as constants

3. Higher-Order Partial Derivatives

Definition 2: Second-Order Partial Derivatives

A **second-order partial derivative** is a partial derivative of a partial derivative. The two *pure* second partials f_{xx} and f_{yy} measure **concavity along coordinate directions** — how fast the slope f_x changes as you move further in x , and similarly for y — exactly analogous to the second derivative of a single-variable function. The two *mixed* partials f_{xy} and f_{yx} measure something new: how the slope in *one* direction changes as you move in the *other* direction, capturing the graph's twist or saddle-like behavior at a point.

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) \quad (\text{pure, in } x)$$

$$f_{yy} = \frac{\partial^2 f}{\partial y^2} \quad (\text{pure, in } y)$$

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \quad (\text{mixed: } x \text{ first, then } y)$$

$$f_{yx} = \frac{\partial^2 f}{\partial x \partial y} \quad (\text{mixed: } y \text{ first, then } x)$$

Theorem 1: Clairaut's Theorem (Equality of Mixed Partial)

If f_{xy} and f_{yx} are both continuous on an open region, then:

$$f_{xy} = f_{yx}$$

The order of differentiation does not matter (under mild continuity conditions).

Why Clairaut's Theorem Works (Proof Outline)

- Consider the second-order difference quotient over a small rectangle $[x, x + h] \times [y, y + k]$
- Define $\Delta = f(x + h, y + k) - f(x + h, y) - f(x, y + k) + f(x, y)$
- Apply the Mean Value Theorem twice: first in x , then in $y \implies \Delta = f_{xy}(c_1, d_1) \cdot hk$
- Apply MVT in the other order: first in y , then in $x \implies \Delta = f_{yx}(c_2, d_2) \cdot hk$
- As $h, k \rightarrow 0$, continuity gives $f_{xy}(x, y) = f_{yx}(x, y)$

When Clairaut Fails – A Counterexample

- Consider $f(x, y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$
- One can show $f_{xy}(0, 0) = -1$ and $f_{yx}(0, 0) = 1$ – the mixed partials are *not equal* at the origin because they are not continuous there

Example 3: Clairaut Verification

Let $f(x, y) = x^3y^2 + e^{xy}$. Find all four second-order partials and verify $f_{xy} = f_{yx}$.

1. First partials:

$$f_x = 3x^2y^2 + ye^{xy}, \quad f_y = 2x^3y + xe^{xy}$$

2. Second partials:

$$f_{xx} = 6xy^2 + y^2e^{xy}$$

$$f_{yy} = 2x^3 + x^2e^{xy}$$

$$f_{xy} = \frac{\partial}{\partial y}[3x^2y^2 + ye^{xy}] = 6x^2y + e^{xy} + xye^{xy}$$

$$f_{yx} = \frac{\partial}{\partial x}[2x^3y + xe^{xy}] = 6x^2y + e^{xy} + xye^{xy}$$

3. Result: $f_{xy} = f_{yx}$ ✓

Example 4: Application: Container Design

A cylindrical container has volume $V(r, h) = \pi r^2 h$ and surface area $S(r, h) = 2\pi r h + 2\pi r^2$. If $r = 5$ cm and $h = 10$ cm, how does V change if we increase r by 0.1 cm? If we increase h by 0.1 cm?

1. Compute partials of V :

$$V_r = 2\pi r h = 2\pi(5)(10) = 100\pi \approx 314 \text{ cm}^3/\text{cm}$$

$$V_h = \pi r^2 = 25\pi \approx 78.5 \text{ cm}^3/\text{cm}$$

2. Estimate changes:

$$\Delta r = 0.1: \Delta V \approx V_r \cdot 0.1 = 10\pi \approx 31.4 \text{ cm}^3$$

$$\Delta h = 0.1: \Delta V \approx V_h \cdot 0.1 = 2.5\pi \approx 7.85 \text{ cm}^3$$

3. Result: Increasing radius has $\sim 4\times$ more effect on volume than increasing height.

Practice Problems

Problem 1. Find f_x and f_y for $f(x, y) = x^2 \cos y + y \ln x$.

Answer: $f_x = 2x \cos y + y/x$, $f_y = -x^2 \sin y + \ln x$

Problem 2. Find all first partials of $f(x, y, z) = xyz + z^2 \sin(x)$.

$$f_x = yz + z^2 \cos x, \quad f_y = xz, \quad f_z = xy + 2z \sin x$$

Problem 3. For $f(x, y) = \arctan(y/x)$, show that $f_{xx} + f_{yy} = 0$ (i.e., f is **harmonic**).

$$f_x = \frac{-y/x^2}{1+y^2/x^2} = \frac{-y}{x^2+y^2}, \quad f_y = \frac{1/x}{1+y^2/x^2} = \frac{x}{x^2+y^2}$$

$$f_{xx} = \frac{2xy}{(x^2+y^2)^2}, \quad f_{yy} = \frac{-2xy}{(x^2+y^2)^2}$$

$$f_{xx} + f_{yy} = 0 \quad \checkmark$$

Problem 4. Use the limit definition to compute $f_x(0,0)$ for $f(x,y) = \begin{cases} \frac{xy}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{0-0}{h} = \boxed{0}$$

Problem 5. Find f_{xxy} for $f(x,y) = e^x \sin(xy)$.

$$f_x = e^x \sin(xy) + ye^x \cos(xy)$$

$$f_{xx} = e^x \sin(xy) + ye^x \cos(xy) + ye^x \cos(xy) - y^2 e^x \sin(xy) = e^x [(1 - y^2) \sin(xy) + 2y \cos(xy)]$$

$$f_{xxy} = e^x [-2y \sin(xy) + x(1 - y^2) \cos(xy) + 2 \cos(xy) - 2xy \sin(xy)]$$